Architecting with AI

From Requirements to Reality

AI doesn't replace architects — it amplifies them.

Chad Green
Senior Systems Architect
Jasper Engines & Transmissions



“I asked AI to write my conference bio.
It called me a *visionary thought leader*
who *disrupts paradigms at scale*.
So… I left it.”

Because honestly? Who’s going to argue with that?

A WORD OF CAUTION

AI is confident. AI is fast.

AI will absolutely write something that sounds incredible

and is completely disconnected from reality.

That's true for bios. And it's very, very true for architecture.

THESIS STATEMENT

> The best architects I know use AI not to replace their thinking, but to accelerate their judgment.

Okay. Now let's ***actually*** talk about how to do that.

Meet the Speaker



Chad Green

Senior Systems Architect

Jasper Engines & Transmissions

35+ YEARS | SOFTWARE ARCHITECT | SPEAKER

ABOUT CHAD

- Software architect with **35+ years** building systems that had to **actually work** in production
- Passionate about making **complex architecture** accessible and **fun**
- Early adopter of AI-assisted development — I've seen **the good, the bad, and the hallucinated**
- Conference speaker, community builder, **occasional over-prompter**
- **Superpower:** Turning **"we need a system that does everything"** into something buildable

Agenda



ACT I

Requirements Are Lying to You

Clarifying requirements with AI, surfacing hidden constraints



ACT II

Design Options, Tradeoffs & Adversarial Insight

Exploring architecture options, stress-testing designs



ACT III

Documentation, Governance & Human Judgment

AI-assisted documentation, avoiding AI slop, human curation

CHAPTER ONE

Act I

Requirements Are Lying to You

Why requirements are never as clear as they look.



Requirements Hide Landmines

Ambiguity
Requirements that mean different things to different people

Contradictions
Conflicting rules nobody noticed during review

Missing NFRs
The performance, security, and reliability expectations that are never written down

Hidden Constraints
Business rules and platform limits that only appear after key decisions are made

Why Humans Miss Things

Cognitive Load

Juggling business rules, edge cases, constraints, dependencies, timelines, and capacity — simultaneously — is impossible.

Anchoring Bias

We latch onto the first interpretation that makes sense and stop exploring alternatives.

Incomplete Mental Models

We can't ask about constraints we don't realize exist.

- *AI doesn't get tired, biased, or anchored.*

AI as a Requirements Clarifier

Raw Requirements → AI Analysis → Structured Insights

Extracts ambiguities

Surfaces the hidden meanings in vague language before they become costly assumptions

Surfaces hidden NFRs

Infers the performance and security requirements nobody wrote down — but everyone expects

Identifies contradictions

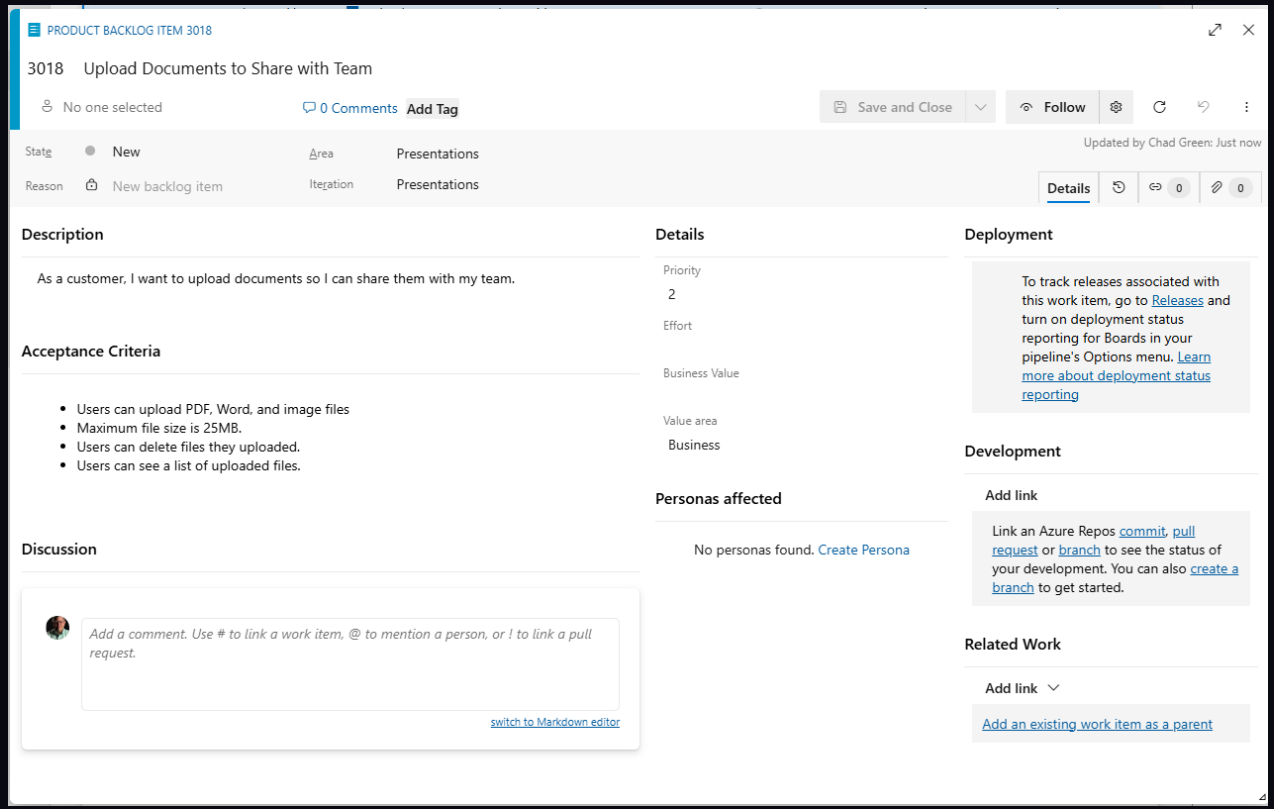
Flags rules that conflict with each other — the ones that survive every review until they break production

Asks clarifying questions

The ones we forget to ask — precisely when it matters most, before design decisions are locked in

AI expands awareness, not assumptions.

Demo Setup: Requirements Clarification



Demo Setup: Requirements Clarification

What You'll See

AI analyzing user stories
Ingesting raw requirements as written

AI surfacing ambiguities & contradictions
Hidden meanings and conflicting rules exposed

AI identifying missing NFRs
Performance, security & reliability gaps

AI generating clarifying questions
The ones the team forgot to ask

The Prompt

copilot-prompt.txt

```
> Analyze the following user story and acceptance
criteria. Identify:

- Ambiguities
- Contradictions
- Missing acceptance criteria
- Implicit non-functional requirements
- Hidden constraints
- Assumptions the stories made
- Questions an architect should ask before designing
  a solution

DO NOT invent new features. Only analyze what is
present or implied.

Ask clarifying questions as you work.
```

LIVE DEMO 1

Requirements Analysis with Copilot

What AI Got Right / What AI Got Wrong

What It Got Right

- Surfaced real ambiguities in the requirements
- Identified missing acceptance criteria no one had flagged
- Pointed out hidden NFRs that matter in production
- Exposed contradictions nobody had noticed

vs

What It Might Get Wrong

- Hallucinated constraints that didn't exist
- Made assumptions not present in the requirements
- Confidently stated incorrect conclusions
- Over-applied patterns from unrelated domains

AI expands the problem space — and the nonsense space. Our job is to separate the two.



AI expands the problem space —
but architects decide what’s real.

AI helps you see more. AI helps you ask better questions.
The judgment is still yours.

SECTION II OF III

Act II

Design Options, Tradeoffs & Adversarial Insight

Exploring more possibilities, faster —
then stress-testing everything.



AI WIDENS THE SOLUTION SPACE

AI for Architecture Exploration

- **Generate 4–5 viable architecture options in seconds** — not just the obvious one
- **Compare tradeoffs** across reliability, scalability, cost, and complexity
- **Evaluate quality attributes** — performance, security, maintainability, observability
- **Explore architectural patterns** you might not have considered

AI doesn't anchor. That's the point.

Demo Setup: Architecture Options

What You'll See

- AI proposing 4 architecture approaches
- AI comparing strengths and weaknesses
- AI identifying risks per option
- AI surfacing hidden assumptions

The Prompt

copilot-prompt.txt

```
> Given the clarified requirements, propose 4
  distinct architecture approaches:

  - Modular monolith
  - Microservices
  - Event-driven
  - Serverless

  For each, include:

  - High-level description
  - Strengths
  - Weaknesses
  - Key risks
  - Team skill assumptions
```

LIVE DEMO 2

Architecture Options + Tradeoffs

Switching to IDE…

The Power of Adversarial AI

After choosing a preferred architecture — stress-test it.

1

Different Model

Use a separate AI perspective to avoid confirmation bias.

2

Different Perspective

Explicitly instruct it to attack, not validate, your design.

3

Brutally Honest

It will surface risks, gaps, and failure modes without sugarcoating.

4

Finds Blind Spots

Hidden assumptions, scaling risks, security gaps, data consistency issues.

⚠ ***The closest thing to a pre-mortem without deploying to production.***

Demo Setup: Adversarial Agent

What You'll See

- AI attacking the chosen architecture
- AI identifying hidden assumptions
- AI surfacing scaling and operational risks
- AI predicting likely failure modes

The Prompt

copilot-prompt.txt

```
> Given the clarified requirements, propose 4
  distinct architecture approaches:

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  - Microservices
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  - Weaknesses
  - Key risks
  - Team skill assumptions
```

LIVE DEMO 3

Adversarial Review

Switching to IDE…

ACT II — KEY TAKEAWAY



AI challenges us in ways humans rarely do — and that makes our designs stronger.

AI gives you breadth. Adversarial AI gives you depth.

Together they build more resilient, defensible architectures.

Act III

Documentation, Governance & Human Judgment

Where AI ends and human judgment begins.

⚠ CAUTION

The Danger of AI Slop

Volume Without Clarity

AI can generate **too much** — volume without clarity is noise, not documentation

Teams Drown in Docs

When **everything** is documented, **nothing** is documented

Architects Must Curate

We decide what **matters**, what **belongs**, and what gets **cut**

⚠ ***AI is a drafting engine — not a decision-maker.***

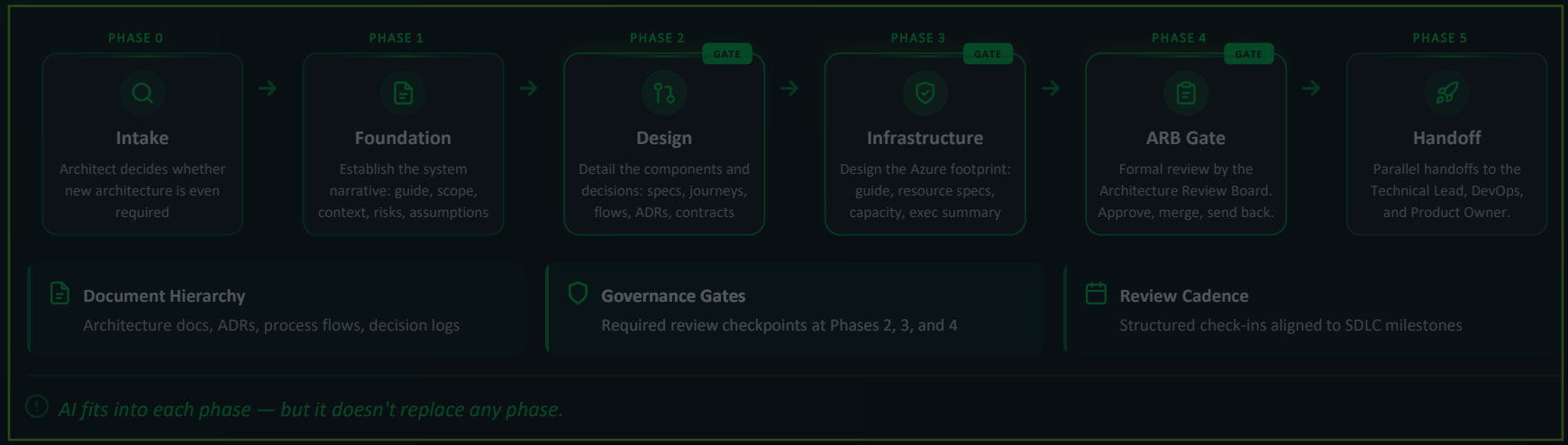
The Jasper Architecture Process



Document Hierarchy

Architecture docs, ADRs, process flows, decision logs

The Jasper Architecture Process



Governance Gates

Required review checkpoints at Phases 2, 3, and 4

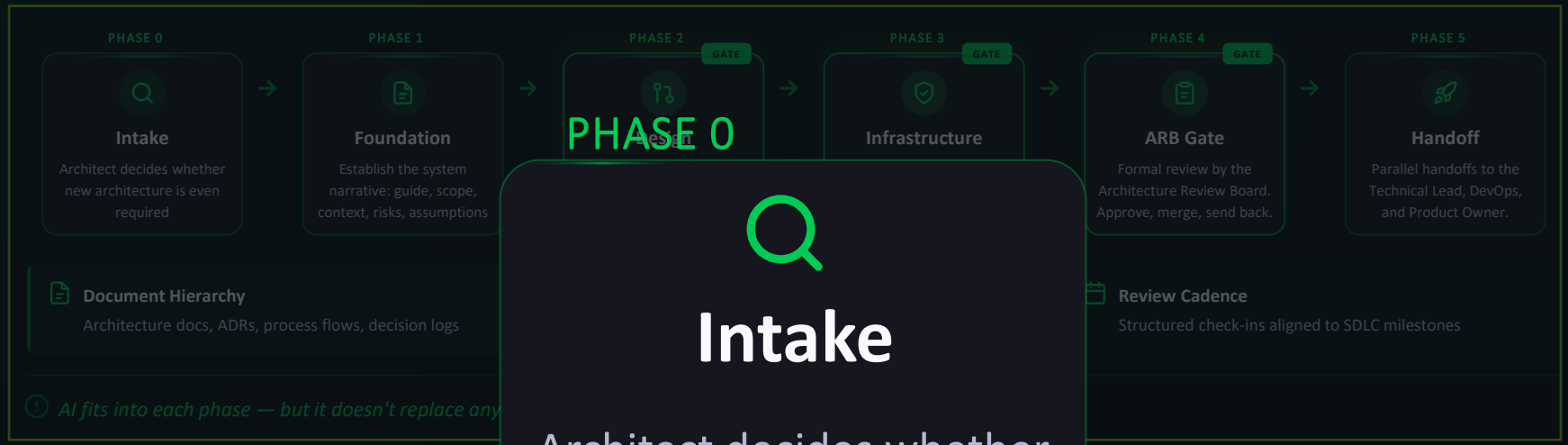
The Jasper Architecture Process



Review Cadence

Structured check-ins aligned to SDLC milestones

The Jasper Architecture Process

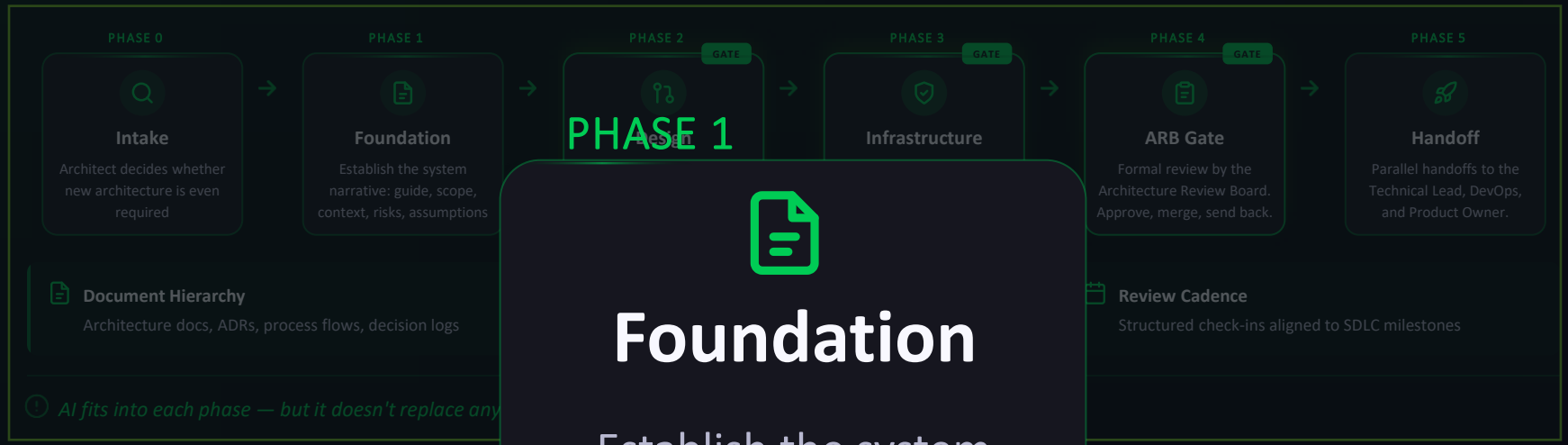


PHASE 0

Intake

Architect decides whether new architecture is event required.

The Jasper Architecture Process

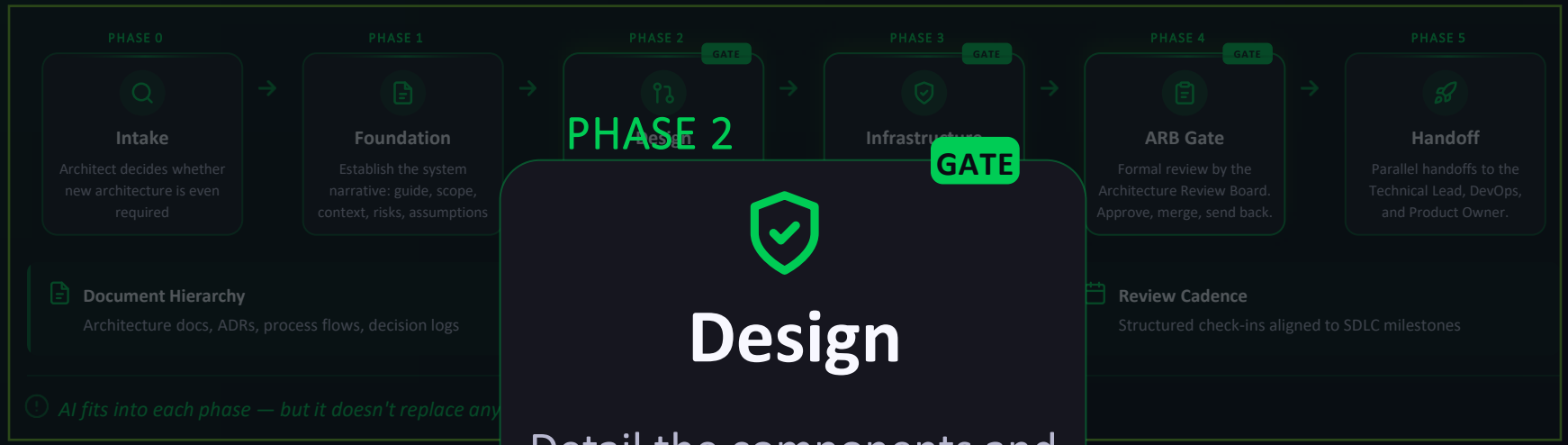


PHASE 1

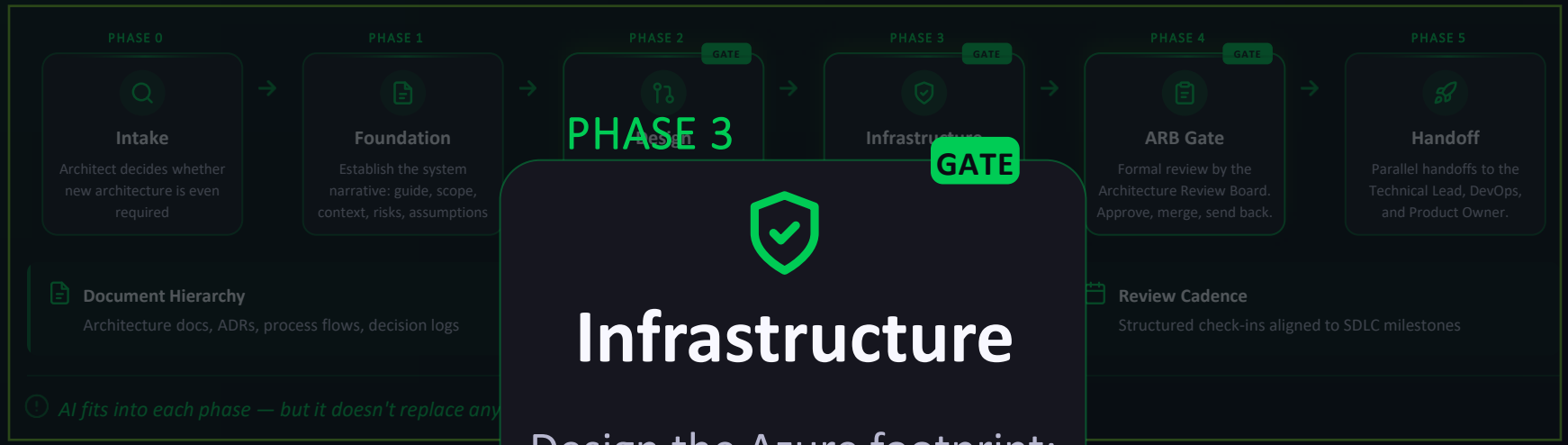
Foundation

Establish the system narrative: guide, scope, context, risks, assumptions.

The Jasper Architecture Process



The Jasper Architecture Process



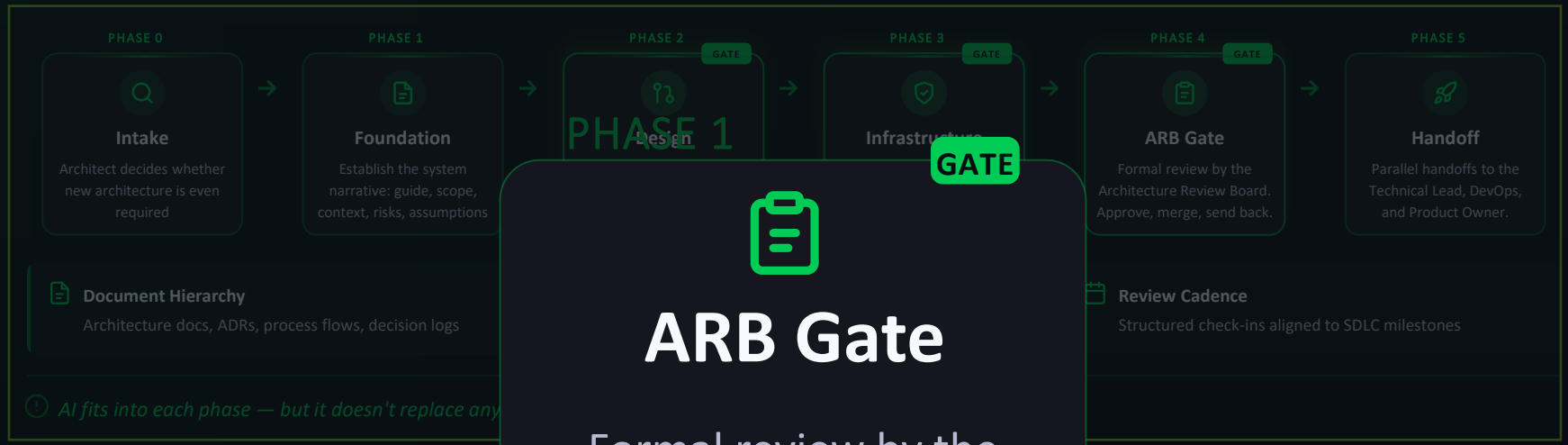
PHASE 3

GATE

Infrastructure

Design the Azure footprint: guide, resource specs, capacity, exec summary.

The Jasper Architecture Process

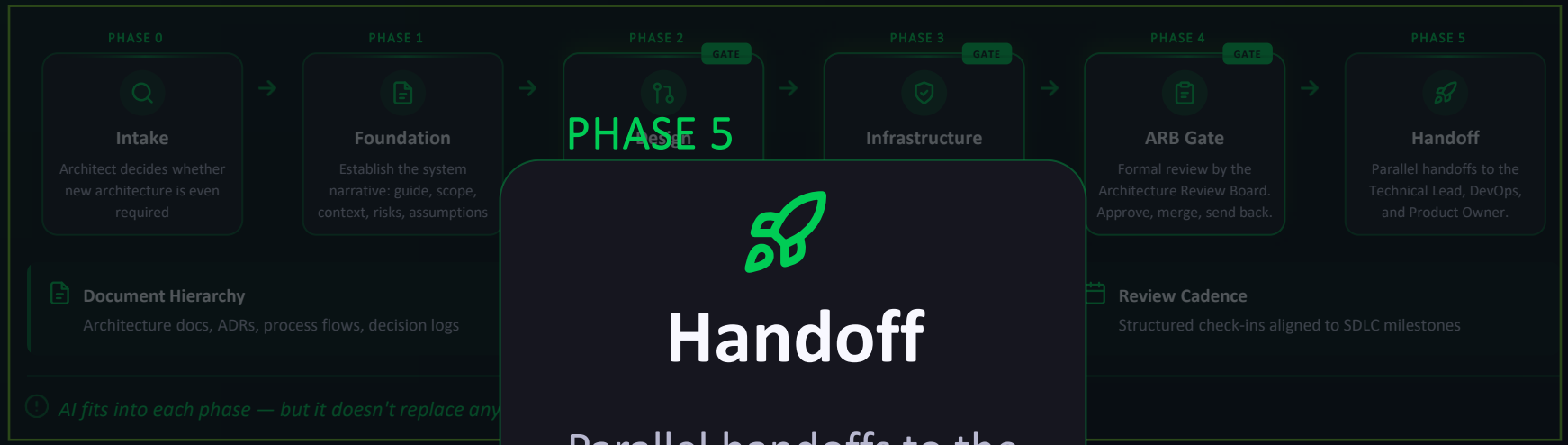


GATE

ARB Gate

Formal review by the Architecture Review Board. Approve, merge, send back.

The Jasper Architecture Process



PHASE 5

Handoff

Parallel handoffs to the Technical Lead, DevOps, and Product Owner.

The Jasper Architecture Process



Document Hierarchy
Architecture docs, ADRs, process flows, decision logs

Governance Gates
Required review checkpoints at Phases 2, 3, and 4

Review Cadence
Structured check-ins aligned to SDLC milestones

AI fits into each phase — but it doesn't replace any phase.

Templates + Instructions + Examples = Quality

Structure → **Clarity** → **Quality**

Prevent AI Slop

Constrain → Focus

Enforce Structure

Ground → Govern

Accelerate Drafting

Speed → Quality

"AI wrote this" vs. **"AI helped me write this"** — the difference is **structure.**

Demo Setup: Document Generation

What You'll See

- ADR generation using a structured template
- Process flow generation from an architecture decision
- How instructions enforce quality and prevent drift
- How unconstrained AI produces slop — contrast example

Structure + instructions = quality output — every time.

The Prompts

ADR PROMPT — TEMPLATE-BOUND

```
> Using the ADR template, instructions, and example (located at
docs-templates/adr), generate an ADR for the decision to use a
[selected architecture].
```

PROCESS FLOW PROMPT — TEMPLATE-BOUND

```
> Using the Process Flow template, instructions, and example
(located at docs-templates/process-flow), generate a process flow
document for the file upload workflow.
```

LIVE DEMO 4

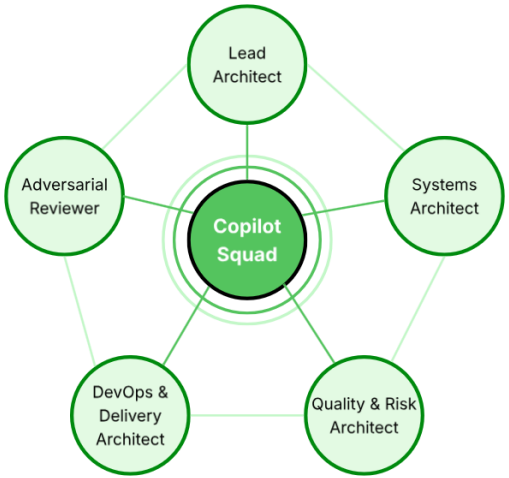
ADR + Process Flow Generation

Switching to IDE…

Copilot Squad

PREVIEW ONLY

Multi-Agent Architectural Workflows



Multi-agent workflows

Specialized AI agents working in parallel — not one model doing everything

Each agent has a distinct role and defined scope

Specialization prevents context bleed and hallucination drift

No single agent holds too much context

Bounded scope keeps each agent focused, accurate, and auditable

Together, they mirror a full architecture review team

Requirements → Design → Tradeoffs → Adversarial Review → Documentation

Seek me out, and I talk all day about Squad.

ACT III — KEY TAKEAWAY

“

AI accelerates drafting — but architects ensure correctness.

AI helps you write faster. AI helps you structure your thinking.
But the judgment, the decisions, the governance — that's still ours.

Closing

Bringing It All Together

Where AI ends and architects begin.

The New Role of the Architect

01

AI Expands Possibilities

More options, more patterns, more tradeoffs, more risks explored — faster than ever before.

02

Humans Provide Judgment

We decide what is real, what is feasible, what is responsible — and we own the consequences.

03

The Partnership Matters

Human judgment plus AI acceleration — not replacement, not automation. Augmentation.

Not replacement. Not automation. Augmentation.

CLOSING STATEMENT

You now have superpowers.

Use them to build better systems, better teams,
and better futures.

Because AI doesn't replace architects — AI amplifies architects.

Thank You

Q & A

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